

Demonstration of Digital Twin Development and Validation Workflows in Heterogeneous Multi-Party Environments using OTDS

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Abstract We demonstrate a Digital Twin (DT) data space enabling policy-governed exchange of Internet Engineering Task Force (IETF)-defined DT components across organizations. The demo showcases interoperable and real-time cross-organizational DT workflows using heterogeneous optical network environments. ©2026 The Author(s)

Review

Digital twins (DTs) have become a key enabler for optical network operation, predictive maintenance, telemetry analytics, infrastructure orchestration, and autonomous network management. The Internet Engineering Task Force (IETF) defines a DT as a virtual representation of the physical network used to analyse, diagnose, emulate, and control network behaviour [1], comprising four elements: data, mapping, models, and interfaces [1]. In optical networks, DT research has mainly focused on physical-layer modelling [2], machine learning (ML)-enhanced models for components [3]–[7], and operational tasks including launch-power optimization, wavelength management, and safe control evaluation [8], [9].

Despite this progress, most DT studies rely on proprietary models, lab environments, or operator-specific datasets, limiting generalizability and reproducibility across heterogeneous optical

infrastructures. A major challenge is enabling governed exchange of DT elements between organizations, such that joint development and validation of DT models can be carried out while preserving sovereignty and confidentiality.

Our prior work introduced the Optical Testbed Data Space (OTDS) [10], a sovereign and policy-enforced cross-stakeholder data sharing platform for trusted collaboration between research testbeds and vendors [11]. Here we extend the OTDS beyond data sharing to a DT data space for optical networks, enabling cross-organizational DT workflows across heterogeneous infrastructures, as shown in Fig. 1. The IETF-defined DT elements, particularly the elements *data* and *model*, become sovereign assets that can be advertised, discovered and exchanged across organizations.

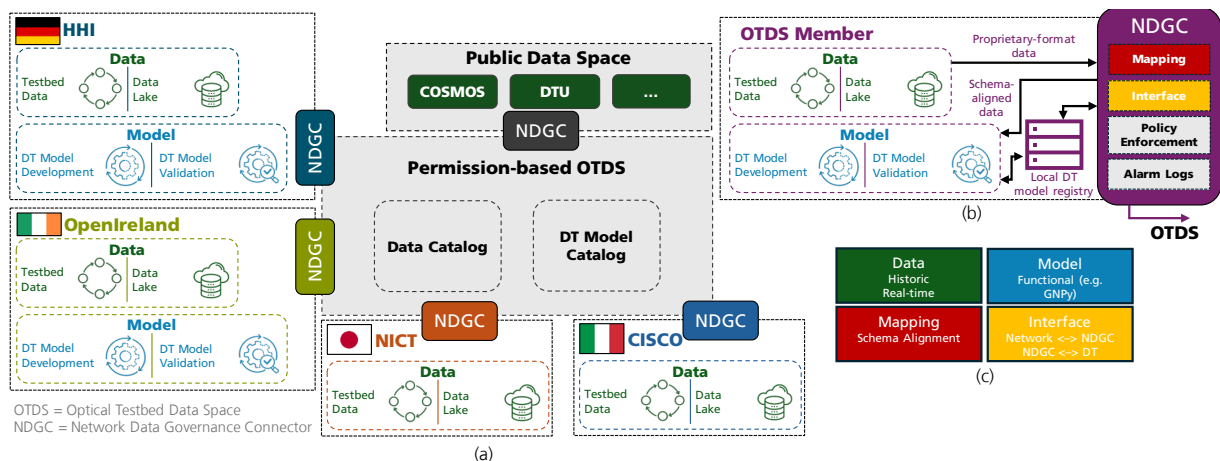


Fig. 1: Cross-organization DT data space architecture: a) four participating sites exposing data and DT model development as assets through NDGC, b) internal per-site structure for an OTDS member that is a provider and consumer, and c) colour legend mapping the IETF DT elements to NDGC functions.

Novelty

The proposed demo extends OTDS to a cross-organization DT data space enabling sovereign and policy-enforced DT workflows. DT data and models are treated as governed assets exchanged across organizations, while DT interface and mapping elements are natively integrated into the data space architecture to support interoperability, synchronization, and distributed validation. This introduces new challenges in cross-organizational schema alignment, DT model governance, and integration of DT assets within a sovereign data-sharing environment.

At the core of the proposed work is the integration of DT environments with the OTDS, enabling governed exchange of data assets, metadata, and DT models under strict policy enforcement, while preserving operational sovereignty, confidentiality, and compliance. Heterogeneous telemetry collected through gNMI, NETCONF, and YANG-based models [12]-[14] is mapped with centralized ecosystem-level semantic mapping aligned with data space vocabulary to support interoperable DT operation without modifying of local infrastructures. The framework combines the four DT ecosystem components defined by IETF [1], in a unified policy-governed environment supporting DT synchronization and validation as shown in Tab. 1. Each participant defines and enforces its own policy set according to organizational and regulatory requirements, ensuring that data and DT assets are shared only when pre-defined conditions are satisfied.

The proposed system supports interactive multi-stage DT workflows, including model exchange, schema harmonization, data exchange, and live DT model validation across geographically distributed environments. The demo showcases these capabilities through an optical-amplifier DT use case using heterogeneous telemetry datasets and distributed DT assets across multiple optical network environments. The optical amplifier is selected because it is a critical optical-network element for DT modelling, has been widely studied, and is highlighted in the OFC 2026 ML Challenge [15]. The work illustrates how sovereign data spaces can provide a practical foundation for scalable, reproducible, and transferable DT deployment in optical networks.

ECOC

Reproducible DT research for optical networks is held back by the scarcity of representative datasets and absence of an interoperable mechanism for sharing DT models across organizations. The proposed OTDS-based DT framework directly addresses ECOC themes in DT development, telemetry analytics, ML for optical

Tab. 1: Comparison of the four IETF DT elements in a conventional single-testbed deployment vs. the proposed DT data space.

IETF DT Elements	DT	DT Data Space (this work)
Data	Single-stakeholder data	Cross-stakeholder data
Model	Single-stakeholder asset without sovereignty features	Governable asset for cross-stakeholder development and testing in a sovereign way
Mapping	Stakeholder proprietary schema	Natively integrated with data space vocabulary for automated policy enforcement during mapping to a unified DT model
Interface	REST / NETCONF / gNMI	REST / NETCONF / gNMI over data space connector

networking, and interoperable network automation, while contributing towards ongoing standardization efforts related to DT interoperability and sovereign optical data spaces. At ECOC we intend to use the demo to gather community feedback on schema design, policy definition, and adoption barriers, as well as onboard new participants into the DT data space.

Demo Implementation

This demo shows how OTDS can serve as the foundation for cross-organizational DT modelling and validation in optical networks. The proposed setup supports sovereign dataset sharing, DT model exchange, schema alignment across heterogeneous infrastructures, and DT model validation. As shown in Fig. 1, the ecosystem comprises two data spaces: permission-based OTDS and a public data space. The permission-based has four authorized members: Fraunhofer HHI, NICT, OpenIreland, and Cisco, while the public data space offers open-source optical datasets [16][17] that can also be exchanged with OTDS. Each participant hosts a Network Data Governance Connector (NDGC) – an Eclipse Dataspace Components Connector (EDCC)-based instance customized for optical networks, local data repositories and DT assets, with the DT architecture follows the IETF model. In this setup, HHI acts as a data consumer and DT developer, NICT and Cisco primarily as data and DT asset providers, and OpenIreland as both provider and consumer. To reconcile heterogeneous telemetry formats, a unified DT schema is defined, while mapping and transformation scripts within the NDGC align local datasets to this schema before exchange. DT models are treated as sovereign data assets that can be advertised, discovered, exchanged, and

deployed across organizations under the same governance rules as datasets.

The demo scenario comprises five heterogeneous optical testbeds natively connected to OTDS: one from Cisco, one from NICT, one from OpenIreland, and two from Fraunhofer HHI. Cisco contributes a laboratory testbed based on a DWDM C+L-Band Open Line System, spanning 12x75km ELEAF fiber spans with full channel loading at 150 GHz spacing. It uses real 800G MPCS 138 GBaud coherent signals together with channelized ASE loading. The data include plug-gable power monitor outputs, amplifier telemetry, channel power distributions, spectral profiles, amplifier gain characteristics, and end-to-end optical performance metrics. NICT contributes the Koganei long-distance optical transmission testbed, comprising ~1200 km optical relay transmission over 12 spans of 100.4 km single-mode fiber, 56 wavelength channels on a 75 GHz grid, ROADM-based infrastructures, and spectrum analysers at intermediate nodes. OpenIreland contributes EDFA profiling datasets and amplifier characterization information supporting transfer learning workflows. Pre-collected datasets are used whenever possible, with additional data generated on demand. Fig. 2 illustrates these setups. HHI offers a 3-span transmission setup presented in [18] and a 6-ROADM node commercial testbed presented in [19][20]. The end-to-end DTs of these testbeds host the DT model exchanged over the data spaces.

Several workflows for cross-organization DT model development and validation will be presented at ECOC. One exemplary workflow, focused on cross-organizational EDFA DT validation, is shown in Fig. 3. Cisco, NICT,

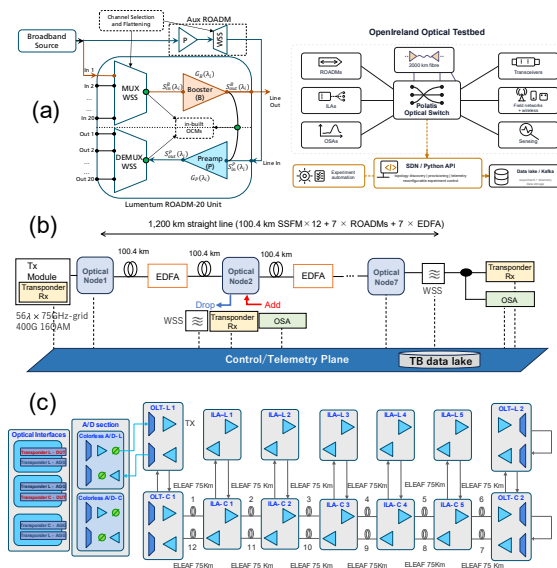


Fig. 2: Testbed setups: (a) OpenIreland (C-band); (b) NICT (C-Band); (c) Cisco (C+L-Band)

OpenIreland, and open-source repositories advertise EDFA datasets and DT assets through OTDS together with access and usage policies. HHI discovers and requests datasets and/or DT models. If policies are satisfied, transformation scripts convert heterogeneous datasets into unified schema, and the data and/or models are exchanged. HHI then validates the DT models with inference executed during the live demo. All exchanges are governed by the NDGC policy enforcement engine, preserving data sovereignty, protecting vendor confidentiality, and ensuring compliance with organizational requirements.

Visitors will engage with the setup through dedicated dashboards and interfaces, view data and DT model catalogues, request assets, observe policy enforcement outcomes, trigger DT workflows, compare predictions across testbeds, and observe DT adaptation. Dashboards will visualize EDFA gain spectra, channel loading conditions, amplifier telemetry, DT predictions, and validation outputs. By stepping into the role of a vendor, testbed owner, or compliance officer, visitors can observe first-hand how sovereign data spaces enable collaborative DT development across organizational and regulatory boundaries.

Conclusions

This demo extends the OTDS from sovereign data sharing to sovereign exchange of DT assets and models, enabling DT validation workflows spanning HHI, NICT, OpenIreland, Cisco and open-source repositories. DT models are treated as governed assets, and along with the schema alignment mechanisms introduced in this work, enable collaborative DT development across organizational boundaries, supporting large-scale DT operations across heterogeneous and distributed optical environments.

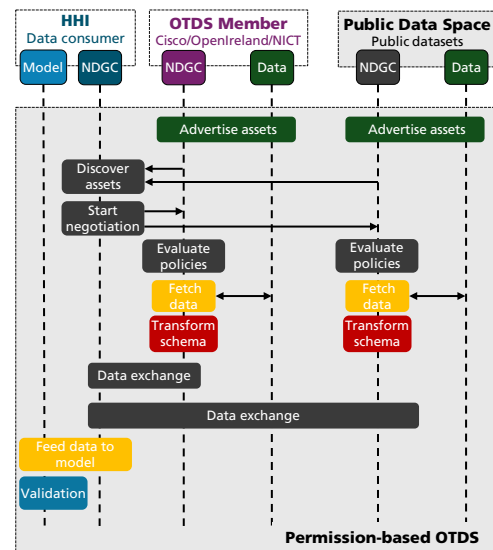


Fig. 2: One exemplary workflow of the demo.

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